# Lightweight Temporal Deep Learning for Real-Time American Sign Language Recognition

***A Dual-Model Pipeline Using MediaPipe Landmarks, MLP, and BiLSTM***

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## 1 INTRODUCTION

- 70M deaf/hard-of-hearing people worldwide use sign language as primary communication
- ASL is the most common visual language for deaf people in North America
- Real-time ASL recognition is challenging: computational cost, lighting sensitivity, diverse users
- This thesis proposes a lightweight real-time ASL system using MediaPipe + MLP + BiLSTM on standard CPU

## 2 RESEARCH QUESTIONS

RQ1 How effectively can a lightweight landmark-based pipeline achieve real-time ASL recognition without compromising accuracy?
RQ2 What is the optimal balance between model complexity and performance in hybrid static-dynamic gesture recognition?
RQ3 How does MediaPipe landmark detection compare to traditional CNN approaches in computational efficiency?

## 3 OBJECTIVES

- Review existing SLR systems; collect custom static (34 classes) + dynamic (12 classes) dataset
- Build MLP for static + BiLSTM for dynamic gesture recognition with auto mode switching
- Evaluate: accuracy, precision, recall, F1, latency, FPS; assess GDPR/EU AI Act compliance

## 4 METHODOLOGY

- Design Science Research: problem analysis, literature review, system design, implementation, evaluation
- Dual-model: MLP (spatial) + BiLSTM (temporal), privacy-preserving with 63-D landmark arrays only
- Stack: Python 3.11, TensorFlow 2.21, MediaPipe 0.10.14, OpenCV 4.x, Streamlit

## 5 SYSTEM ARCHITECTURE

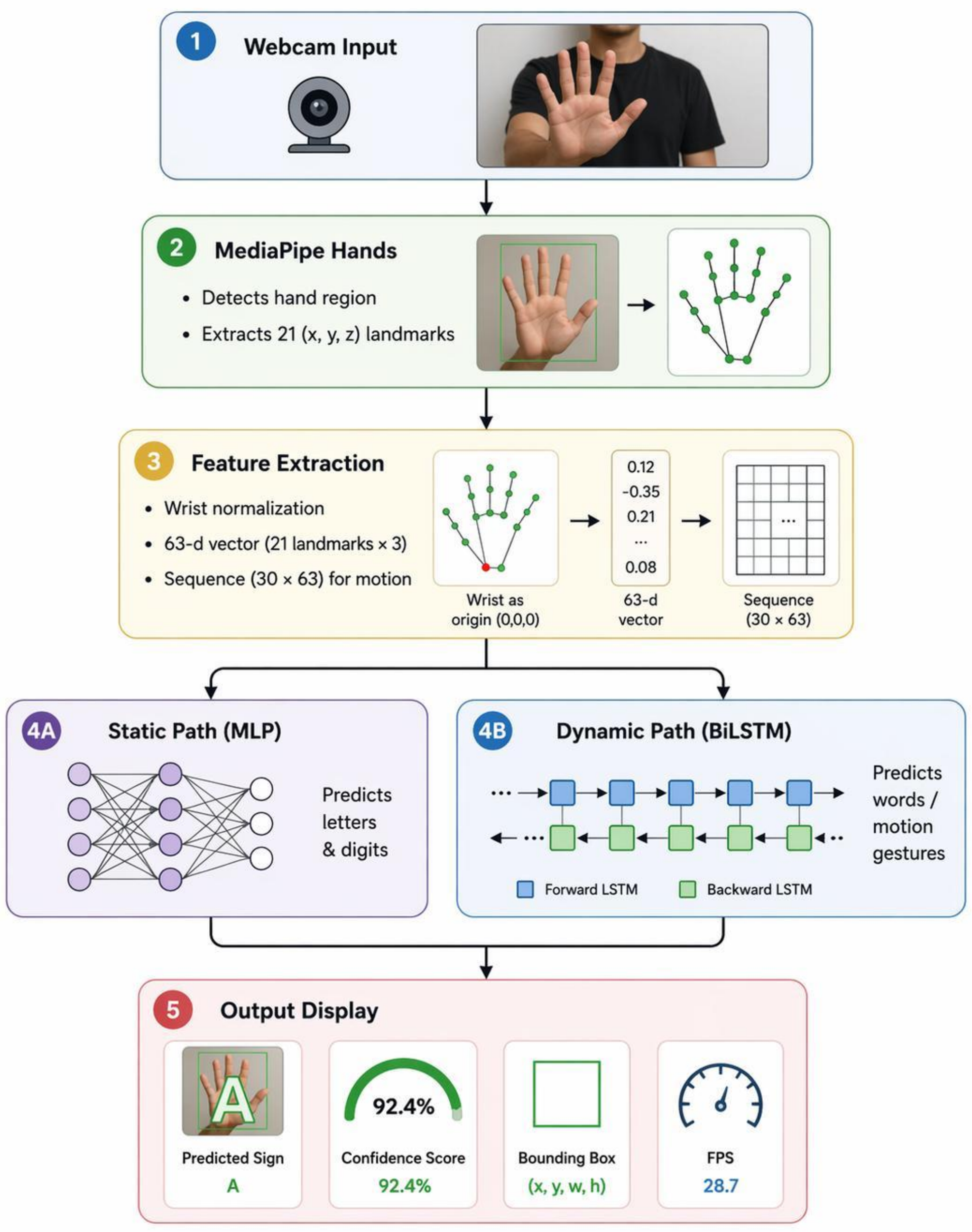
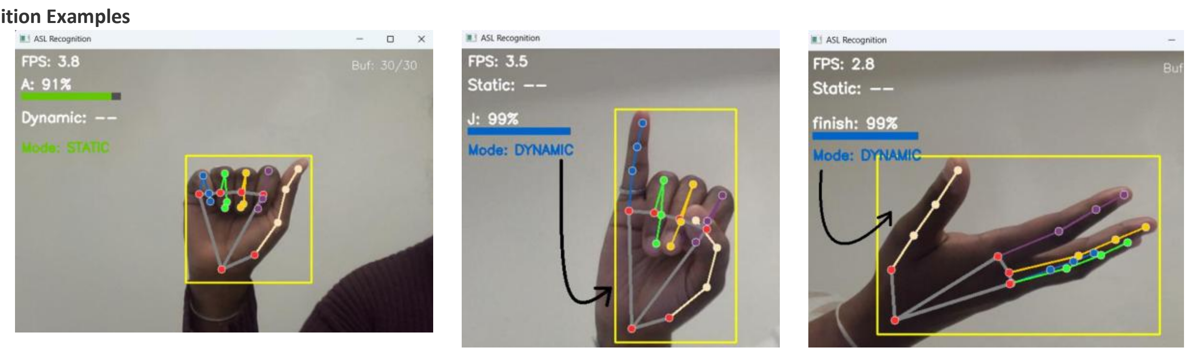



*Figure 1. System pipeline*

**Live Recognition Examples**




A: 91%    J: 99%    finish: 99%

*Figure 2. Real-time recognition: static (A) and dynamic (J, finish)*

## 6 MODEL ARCHITECTURE & TRAINING

- Static MLP: 63→128(ReLU,Drop0.3)→64(ReLU,Drop0.3)→32(ReLU)→34(Softmax)
- ~19,650 parameters | 268 KB | Adam lr=0.001, batch=32, epochs=100, early stopping patience=15
- Dynamic BiLSTM: (30,63)→BiLSTM(64)→BiLSTM(32)→Dense(64,ReLU)→Drop(0.5)→12(Softmax)
- ~111,628 parameters | 1.37 MB | Adam lr=0.001, batch=32, epochs=150, early stopping patience=20

**Mode-Switching Mechanism**

- Tracks wrist landmark displacement over 6 frames using Euclidean distance
- Average displacement > 0.012 normalised units → dynamic mode
- Hysteresis: 20 consecutive still frames before returning to static

**Key Design Decisions**

✓ Separate models per modality (spatial vs. temporal)
✓ Common 63-D representation eliminates dual preprocessing
✓ Zero-cost mode switching at feature level
✓ Total < 2 MB, no GPU required

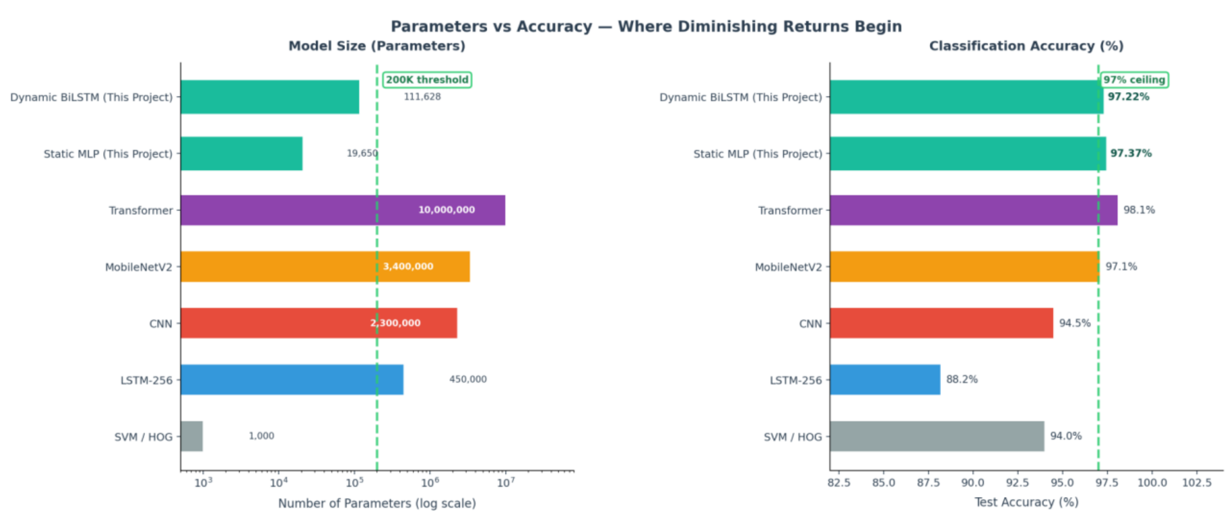



*Figure 4. Parameters vs. accuracy tradeoff*

## 7 DATASETS & COLLECTION

- Custom dataset via webcam with MediaPipe-based interface, indoor setting
- Static: 3,300 samples, 34 classes (A-Y + 0-9), split 2,309/496/495
- Dynamic: 1,200 sequences, 12 classes (J,Z + 10 words), split 840/180/180

**MediaPipe vs. CNN Approach (RQ3)**

| Aspect | MediaPipe (Ours) | CNN Baselines |
|---|---|---|
| Input Size | 63 features | 921,600 pixels |
| CPU FPS | 10.6 FPS | 2.5-4.0 FPS |
| Privacy | Landmarks only | Raw images stored |
| Hardware | Any webcam | GPU often needed |

## 8 INFERENCE SPEED & COMPLEXITY

- MediaPipe: 10.6 FPS on CPU vs. CNN 2.5 FPS, MobileNetV2 3.0 FPS, LSTM-256 4.0 FPS

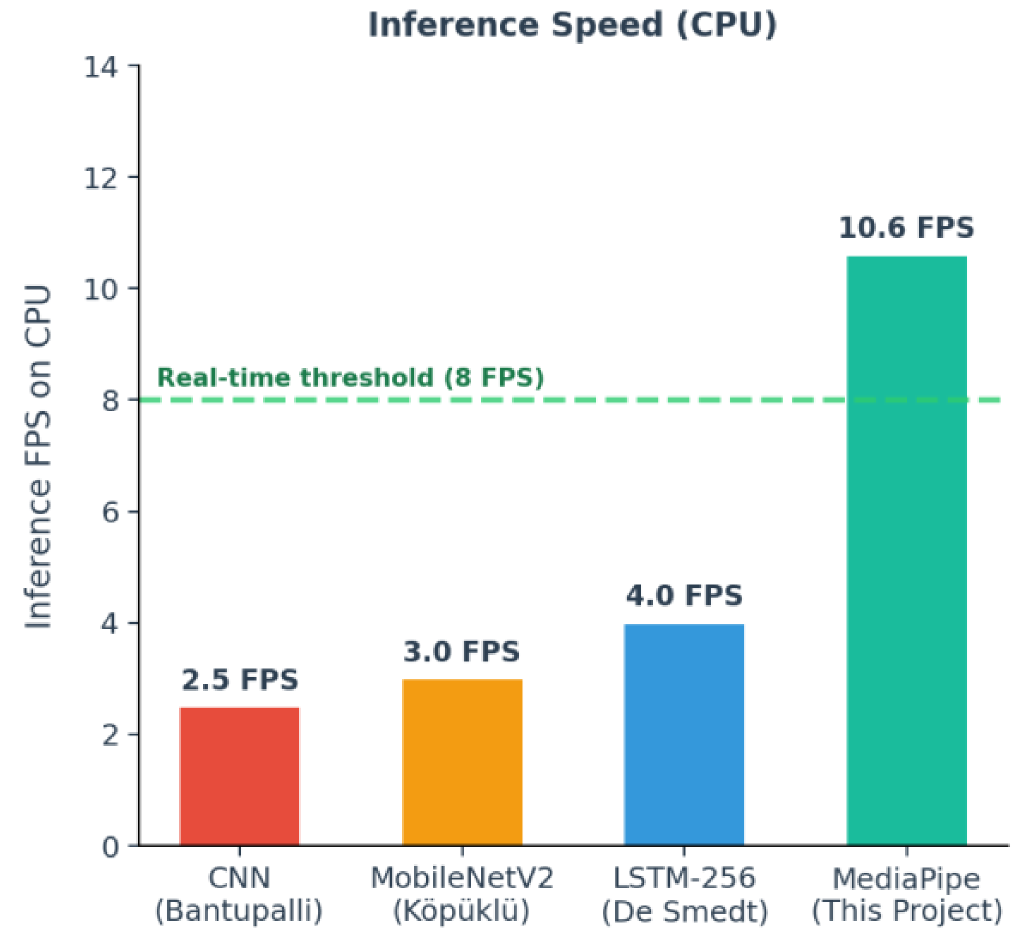



*Figure 3. CPU inference speed comparison*

## 9 PERFORMANCE METRICS

| | |
|---|---|
| **97.37%** Static MLP Accuracy | **97.22%** Dynamic BiLSTM Accuracy |
| **37 FPS** Hand Detection Speed | **~120ms** End-to-End Latency |

**Classification Performance**

| Metric | Static MLP | Dynamic BiLSTM |
|---|---|---|
| Accuracy | 97.37% | 97.22% |
| W. Precision | 97.69% | 97.35% |
| W. Recall | 97.37% | 97.22% |
| Macro F1 | 97.29% | 97.22% |
| Model Size | 268 KB | 1.37 MB |
| Parameters | ~19,650 | ~111,628 |

## 10 KEY FINDINGS

RQ1 97.37% / 97.22% accuracy at 37 FPS on standard CPU – all targets exceeded
RQ2 Dual-model outperforms single model; total < 2 MB with auto mode switching
RQ3 MediaPipe reduces input 14,629:1 vs. raw pixels, CPU-only deployment

- Most confused: U→R (4), 0→O (4), J→finish (2) – visually similar gestures

**Confusion Matrix Analysis**

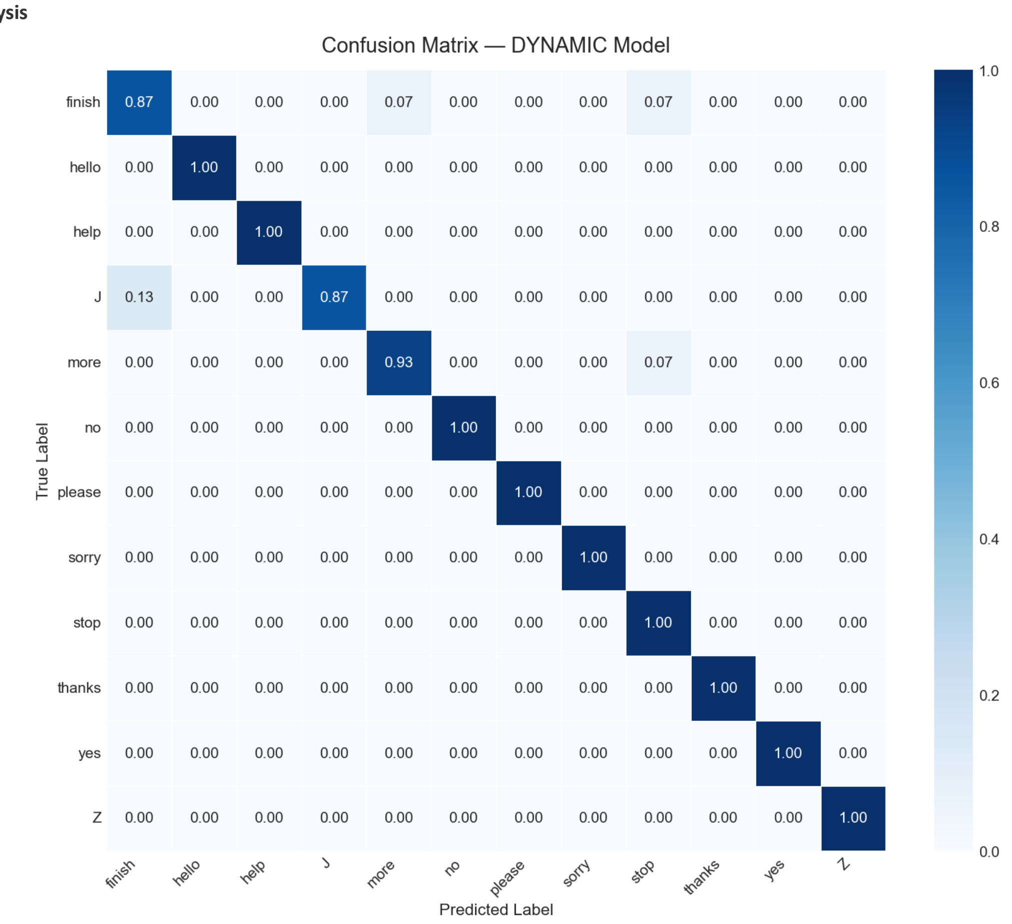



*Figure 5. Dynamic model confusion matrix (12 classes)*

## 11 CONCLUSIONS & FUTURE WORK

- Real-time ASL feasible on commodity hardware; both models exceed 95% target
- Privacy-preserving: only landmarks stored, no raw video/images
- 37 FPS detection, ~120ms latency on standard CPU
- Future: full fingerspelling, two-hand gestures, TFLite mobile, text-to-speech

## 12 REFERENCES

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